
An Audit of Structure-Based HIV Drug-Resistance Triage and Its LLM Explanations

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Abstract

1 Antiviral drug resistance forces medicinal chemists to triage candidate inhibitors
2 by how well their binding survives known resistance mutations. We present **Re-**
3 **sistScope**, a target-agnostic pipeline that docks a candidate against a wildtype
4 receptor and a panel of clinical resistance mutants, scores binding robustness
5 ($\Delta\Delta G$), and writes a per-mutation mechanistic explanation with an LLM. Both
6 outputs have real ground truth, so we audit both. On HIV-1 protease (6 PIs) and
7 reverse transcriptase (4 NNRTIs) against Stanford HIVdb data, **the docking score**
8 **is beaten by its own geometry**: ranking mutations by distance to the ligand, same
9 receptor but no docking run, tracks measured fold-resistance better than $\Delta\Delta G$ (ρ
10 0.39 vs. 0.00 on protease, above the HIVdb expert system’s 0.36) and recovers
11 known resistance mutations better on both targets. The energy earns its cost in
12 one place only, at the top of the ranking in the rigid NNRTI pocket, where each
13 drug’s top ten holds 7 to 8 real resistance mutations against 3 to 4 for distance.
14 On the flexible protease active site that ordering reverses. Turning to the explana-
15 tions, an LLM judge scoring against IAS-USA and PubMed-derived mechanisms
16 marks 71–72% as recovering the correct primary mechanism, and an **attribution**
17 **ablation** over real, absent and corrupted structural context shows they condition
18 on the pipeline, most strongly where the model’s prior is weakest. A verification
19 that uses no language model reproduces the same ordering. Across both halves,
20 structural geometry carries the signal and the docking energy mostly does not.

21 1 Introduction

22 Resistance is the central failure mode of antiviral therapy. A medicinal chemist with twenty candi-
23 date inhibitors must decide which few to advance into slow, expensive phenotypic resistance assays,
24 a decision currently made by hand, mutation by mutation. A natural computational aid is to dock
25 each candidate against a panel of known clinical resistance mutants and rank candidates by how well
26 predicted binding *survives* the panel.

27 Such a tool raises two questions that are usually asked separately, and that both have ground truth:

- 28 1. **Does the prediction hold up?** Stanford HIVdb records *measured* fold-resistance for es-
29 sentially every approved HIV drug [19]. This turns “is the score reasonable?” into “does
30 the score predict a measured number, and does it beat the trivial baselines a chemist could
31 use instead?”
- 32 2. **Does the explanation hold up?** If the tool also explains *why* a mutation threatens a com-
33 pound, that explanation can be checked against mechanisms experts have already doc-
34 umented. Most structure-based resistance work asks whether the score matches known
35 biology, not whether the *reasoning* does.

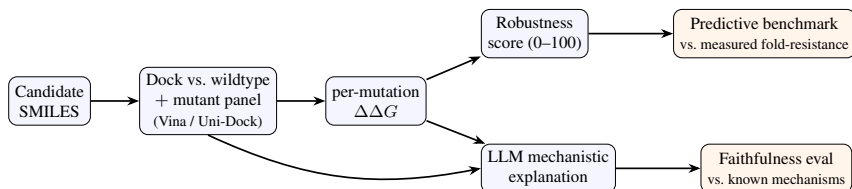


Figure 1: **ResistScope pipeline.** A candidate SMILES is docked against wildtype plus a panel of clinical resistance mutants; per-mutation $\Delta\Delta G$ yields a robustness score and, with structural context, an LLM mechanistic explanation. Each output is validated against ground truth: $\Delta\Delta G$ against measured fold-resistance, explanations against documented mechanisms.

Our central observation is that the structure earns its keep while the *scoring* step mostly does not. Ranking mutations by geometry alone beats the docking energy on both targets, and the explanations lean on that same geometry rather than on the energy. The practical consequence is specific: the expensive part of the pipeline is worth running in one identifiable regime, and a free structural measurement does as well or better in the rest. This is the gap between a retrospective benchmark number and the decision a chemist actually makes.

Contributions. (i) An audit of structure-based resistance triage on two HIV targets, showing a docking-free geometric baseline outranks docking $\Delta\Delta G$ on both, with one exception at the top of the ranking in the rigid NNRTI pocket where the energy does earn its cost. (ii) Decision-relevant evaluation in place of a single pooled statistic: per-drug precision@ k , significance-tested baselines, and a dropout sensitivity analysis. (iii) An attribution ablation for the pipeline’s LLM explanations, transferring the corrupted-context logic of Min et al. [15] to structural input, with a verification that uses no language model. (iv) A reproducible target-agnostic implementation, with all docking results, explanations, ground truth and judge scores released.

2 Related work

Rule-based expert systems are the clinical standard for interpreting HIV resistance genotypes and are our strongest baseline. The Stanford HIVdb penalty algorithm [14, 19] and the IAS-USA mutation lists [22] encode decades of accumulated genotype–phenotype and clinical-outcome evidence into per-mutation, per-drug scores. They are accurate precisely because that evidence exists; they say nothing about a compound or target for which it does not.

Supervised ML on genotype–phenotype data is the dominant computational approach and reaches high retrospective accuracy. *geno2pheno* learns decision trees and SVMs mapping sequence to phenotypic resistance [3]; deep architectures (MLP, bidirectional RNN, CNN) have been trained across 18 antiretrovirals [20]; large-scale gradient boosting and random forests over clinical sequence corpora recover both known and novel resistance mutations [4]; and protein language-model embeddings now push mean AUC to ≈ 0.97 [10]. These are strong, obvious baselines and we do not claim to beat them. They share one requirement that motivates our work: each needs a labeled resistance dataset *for the drug in question*. Docking is training-free, so it applies to a novel scaffold or a target with no resistance corpus, where these methods cannot. Whether that applicability is worth anything is what we test.

Structure-based resistance prediction has a long history and an established explanation for why protease is hard. The substrate-envelope hypothesis [13, 16] holds that protease resistance mutations selectively disrupt inhibitor binding at positions where the inhibitor protrudes beyond the consensus substrate volume, while leaving substrate processing intact. That is a coevolutionary, network-level constraint, and a single-structure pocket-affinity score is not designed to capture it. This predicts our protease result. Other structure-aware approaches include molecular field mapping over homology models [18] and, at the accurate-but-expensive end, alchemical free-energy calculations (88% accuracy on 144 clinical Abl mutations, ≈ 1.1 kcal/mol RMSE [11]) at orders of magnitude more compute per mutation. Receptor flexibility is the standard remedy for rigid-docking failure [2]; we do not employ it, which bounds our negative protease result. Learned co-folding models are a fast-moving alternative [5], though benchmarking shows deep docking can produce physically invalid poses and generalize poorly off-distribution [7].

LLM explanation faithfulness. Whether a model’s stated reasoning reflects the computation producing its answer is distinct from whether the answer is right [12]. Chain-of-thought rationales can be *plausible but unfaithful*, rationalizing answers driven by features the model never mentions [21]. LLM-as-judge evaluation carries known biases that make absolute scores judge-dependent [24], which is why we report cross-model agreement and rest our claims on paired within-judge contrasts. Our corrupted-context condition follows the ablation logic of Min et al. [15], who showed that replacing in-context labels with random ones barely degrades performance, implying the model was not conditioning on them; we apply the same test to structural context. Perturbing the input and re-scoring with an LLM judge is an established family: Porbeck et al. [17] vary debugging context and judge the resulting failure explanations, and Afolabi et al. [1] use causal ablation to test whether stated medical reasoning drives the answer. Our design is not new; its transfer to *structural* input is, as is a verification neither admits, since one claim in every explanation here is checkable against a PDB file with no language model. LLMs are increasingly deployed as tool-using agents in chemistry [6], but the faithfulness of their mechanistic claims in this domain is rarely measured.

3 Methods

Pipeline (Fig. 1). A candidate SMILES is embedded (RDKit/meeko) and docked (AutoDock Vina [9] on CPU, or Uni-Dock [23] on GPU) against a cleaned wildtype receptor and a per-drug panel of point-mutant receptors; the per-mutation $\Delta\Delta G$ (mutant minus wildtype binding) feeds a 0–100 robustness score. Mutant receptors are built by PDBFixer point mutagenesis (Appendix E). The Vina score is an empirical scoring function, not a binding free energy; we use it as a ranking signal only.

Targets and data. *Protease* (PR): PDB 3OXC homodimer, Asp25 dyad asymmetrically protonated; 6 PIs, 285 mutant receptors, 1,591 pairs. *Reverse transcriptase* (RT): PDB 3V81 p66/p51 heterodimer, NNRTI pocket; 4 NNRTIs, 423 mutant receptors, 1,269 pairs (Appendix E). We scope RT to NNRTIs (allosteric pocket binders): rigid docking is a sound proxy for NNRTI affinity but not for NRTI mechanisms. Measured fold-resistance and DRM labels come from Stanford HIVdb / IAS-USA [19, 22].

Explanations and ground truth. For each mutation we assemble a structural-context record (residue identity, Δ volume/charge/hydrophobicity, ligand distance, direct-contact flag, subpocket) plus the docking $\Delta\Delta G$, and prompt an LLM (Claude Haiku 4.5; larger models decline HIV-resistance *generation*, though not judging) for a 2–3 sentence mechanistic hypothesis. Ground truth combines a curated set (IAS-USA + primary literature) with an *agentic* builder that researches each mechanism over live PubMed, cites the papers, and self-verifies grounding. All RT ground truth is agent-built; PR ground truth is hand-curated, so we lead the faithfulness headline with PR.

Faithfulness and attribution. An LLM-as-judge scores each explanation against the ground-truth mechanism on a 0/1/2 rubric; we assess judge robustness with independent models (§4.2). The attribution ablation regenerates explanations under *full* (real context + $\Delta\Delta G$), *minimal* (drug + mutation only), and *corrupted* (pipeline-derived fields deranged across pairs, intrinsic chemistry kept) conditions with the same model and judge; a field-level variant drops one context group at a time.

Predictive benchmark. Every claim carries a permutation p -value or bootstrap 95% CI: top- N DRM enrichment, threshold-free ROC/PR-AUC, a de-confounding analysis (rebuild the fold-resistance target from progressively less co-mutated isolate subsets), a leave-one-drug-out jackknife, and baselines (mutation prevalence, physicochemical $|\Delta$ volume|, and the Stanford HIVdb penalty score via the Sierra GraphQL API). We report many tests without multiplicity correction; the headline contrasts survive at $p < 10^{-3}$ (Appendix E).

4 Results

4.1 The predictive benchmark: geometry, not energy

On both enzymes, rigid single-structure docking- $\Delta\Delta G$ is beaten by a baseline that uses the same receptor but never docks anything. The one exception tells us where the energy step is worth its cost.

Table 1: **Baseline head-to-head.** DRM-recovery ROC-AUC / magnitude ρ vs. measured fold-resistance. Mutation prevalence, $|\Delta\text{volume}|$ and bootstrap CIs are in Appendix C; HIVdb’s ROC is partly circular against the expert DRM labels, so ρ is the fair column.

predictor	protease (ROC / ρ)	RT (ROC / ρ)
docking $\Delta\Delta G$	0.51 / 0.00	0.70 / -0.11 (de-conf. $+0.36$)
—ligand distance (<i>no docking</i>)	0.96 / 0.39	0.80 / 0.13
HIVdb penalty	0.86 / 0.36	0.93 / 0.27

A docking-free geometric baseline beats $\Delta\Delta G$ globally, on both targets. The field-level ablation of §4.3 finds the *explanations* carried by geometry rather than energy; so is the *ranking*. Scoring each mutation by how close its residue lies to the ligand, with no docking run, raises the magnitude correlation with measured fold-resistance from $\rho = 0.002$ to 0.387 on protease, above HIVdb’s 0.360 , and DRM-recovery ROC-AUC from 0.51 [$0.46, 0.56$] to 0.963 [$0.954, 0.971$] (RT: 0.70 to 0.800 [$0.738, 0.854$]). We lead with the magnitude column because DRM labels are enriched for pocket-proximal residues almost by construction, which makes the ROC comparison partly definitional. The ordering is the same on both.

The exception: at the top of the list, on RT, docking wins. ROC-AUC scores a whole ranking, but the tool is read as the top few flagged mutations for one compound, and the two metrics disagree. Taking each drug’s own top 10, docking recovers 7 to 8 genuine clinical DRMs per NNRTI against 3 to 4 for the distance baseline. On protease the result reverses, 0 to 3 for docking against 7 to 10 for distance (Appendix C). In the NNRTI pocket proximity is necessary but not sufficient, since many close residues are not resistance mutations, and discriminating *within* the pocket is the one thing the energy buys. On protease it buys nothing at any depth of the list.

Why the targets differ (Fig. 2, Appendix A). On protease docking sits at chance (Spearman ≈ 0 ; top-40 enrichment $2.86\times$, $p < 0.001$, so tail enrichment coexists with chance-level global ranking) and de-confounding does not rescue it ($\rho \leq 0$). That is expected: protease resistance is co-evolving and substrate-envelope-constrained [13, 16], violating the locality the method assumes (Appendix F). On RT the same protocol is above chance (enrichment $8.35\times$) and de-confounding *does* rescue the magnitude correlation, $-0.11 \rightarrow \rho = 0.36$ [$0.12, 0.56$] ($p = 6 \times 10^{-4}$) on single-mutation isolates.

Why run docking at all, then. Distance and $\Delta\Delta G$ both need only a structure, while the expert and clinical baselines that also beat docking need per-drug data a novel compound lacks (Appendix H); and nothing in the $\Delta\Delta G$ path is fit to resistance phenotypes, so each per-drug number is already a prospective estimate with no training set to hold out. On this evidence the case for docking over a distance filter rests on top-of-list discrimination in a rigid pocket; transfer to a new *target* is untested.

4.2 Explanation faithfulness

Under the Claude-Haiku judge, explanations recover the correct primary mechanism 72% of the time on protease ($n = 46$, hand-curated ground truth) and 71% on RT ($n = 55$, agent-built ground truth, so its absolute value is the weaker of the two); most misses under-specify rather than contradict. The outright misses cluster on distal and inter-subunit DRMs, notably the E138–K101 salt bridge across the p51/p66 interface [8], where the structural context itself is blind rather than the model (Appendix F).

Judge robustness. On a 21-item sample stratified for the judge’s hard (0/1) cases, GPT-5.6 agrees with Claude-Haiku 95% within one point but 48% exactly (κ $0.23/0.38$ unweighted/linear); Claude is systematically *stricter*. Re-grading all 138 protease ablation items with a second Claude model (Sonnet 5) reproduces the §4.3 contrasts (full—minimal $+0.35$, full—corrupted $+0.65$, vs. Haiku’s $+0.43/+0.63$; $\kappa = 0.64/0.72$). We therefore treat the *absolute* percentage as judge-dependent [24] and rest our claims on the paired contrasts.

An out-of-family check. All three judges are LLMs, so the above establishes consistency, not correctness. Appendix G adds a check involving no language model: whether an explanation’s claim about ligand contact matches the receptor geometry. It reproduces the ablation ordering (88% correct under full context, 57% minimal, 55% corrupted; Fisher $p = 5.7 \times 10^{-5}$ full vs. corrupted)

and shows the judge scoring structurally-wrong explanations lower (1.02 vs. 1.51, $p = 6.1 \times 10^{-4}$). Both results most exposed to the circularity objection therefore survive a measurement taken outside the LLM family.

4.3 Attribution ablation: the pipeline drives faithfulness where the prior is weak

Context is genuinely used. Pooled across both targets ($n = 101$), full context beats minimal (drug + mutation only) by $\Delta = +0.27$ ($p = 7 \times 10^{-4}$) and beats corrupted context by $\Delta = +0.46$ ($p < 10^{-4}$). Feeding *wrong* structural facts is worse than feeding none, so by the logic of Min et al. [15] the explanations are conditioning on the pipeline rather than pattern-matching the mutation name (Fig. 3a; Appendices B and D). *How much is the contact flag?* Deranging geometry *within* contact class, leaving that flag true, recovers most of the loss (1.63 vs. 1.15 corrupted, 1.78 full), but real context still wins ($\Delta = +0.15$, $p = 0.035$). The finer geometry contributes beyond the boolean, though the headline arm overstates it fourfold (Appendix E).

Prior-strength dose-response. The benefit of correct context scales inversely with the model’s prior (Fig. 3b). Where the prior alone is *wrong* (minimal = 0, $n = 13$), context corrects the explanation in 11 of 13 cases (85%); where vague (minimal = 1), 66%; where already correct (minimal = 2), context is at ceiling. Because the raw gain is partly mechanical (a 0 has twice the headroom of a 1) we lead with these ceiling-robust rates; restricting to pairs with room to improve (minimal < 2, $n = 45$), context still helps $\Delta = +0.80$ ($p < 10^{-4}$) and the gradient persists (Spearman -0.40 , $p = 0.006$).

Which fields carry the lift. A field-level ablation (protease, $n = 46$; Fig. 3c) drops one context group at a time. Removing the *ligand distance* costs the most ($\Delta = +0.22$, $p = 0.051$) and the *subpocket* next ($\Delta = +0.15$, $p = 0.09$); removing the *docking* $\Delta\Delta G$ ($\Delta = -0.02$) or the physicochemistry ($\Delta = -0.02$) does nothing measurable. Marginally, and on one target ($n = 46$), that points to geometric localization rather than energy. The $\Delta\Delta G$ null is absence of evidence, and specific to *rigid, single-pose, waters-stripped Vina*: a better energy [2, 11] might contribute where ours does not.

5 Discussion

Both halves point the same way: **structural geometry carries the signal, and the docking energy mostly does not.** Ranking by ligand distance beats docking $\Delta\Delta G$ globally on both targets, and dropping distance or subpocket costs the most faithfulness while dropping $\Delta\Delta G$ costs nothing measurable. Two independent analyses reach one conclusion, which is why we report a negative result about the method we built. The energy is not worthless. It discriminates *within* the NNRTI pocket, where proximity alone cannot, and that is the whole of what we can show it buys. Structure still beats the non-structural shortcuts where resistance is structurally local; the *energy* step is what fails to earn its cost. Both boundaries follow from the premise rather than surprising it, and so do the individual failures (Appendix F): eight of nine trace to what the pipeline cannot represent rather than to the language model. We offer this from two contrasting HIV enzymes as a hypothesis, not a law: measure the geometry, treat the docking energy as unproven, and distrust structural triage for flexible, co-evolving sites.

Limitations. Docking-energy claims are bounded by this protocol: rigid single-structure Vina, one pose, waters stripped, and 7.1% of RT dockings dropped for returning no pose. The faithfulness half rests on $n = 46$ and $n = 55$ pairs over two enzymes, with load-bearing contrasts marginal ($p = 0.035$, 0.051 , 0.09) and no multiplicity correction. RT ground truth is agent-built, all three judges are language models, and most of the corruption effect is a single contact flag (§4.3). Appendix E details each.

Future work. A *human-expert anchor*, 20 to 30 explanations graded blind on the same rubric by a structural biologist, is the one measurement that would move the faithfulness result from consistent to correct; the sample and primer are released. *More targets* would test geometry-over-energy beyond two HIV enzymes; neuraminidase and SARS-CoV-2 M^{pro} are already wired in. *A better energy* (ensemble docking [2], free-energy calculation [11]) would separate a null about docking from a null about this protocol. *Replication at larger n*, including on RT, would settle the marginal decomposition claims ($p = 0.035$ to 0.09 at $n = 46$); every number here is also retrospective (§4.1).

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A Predictive benchmark figures

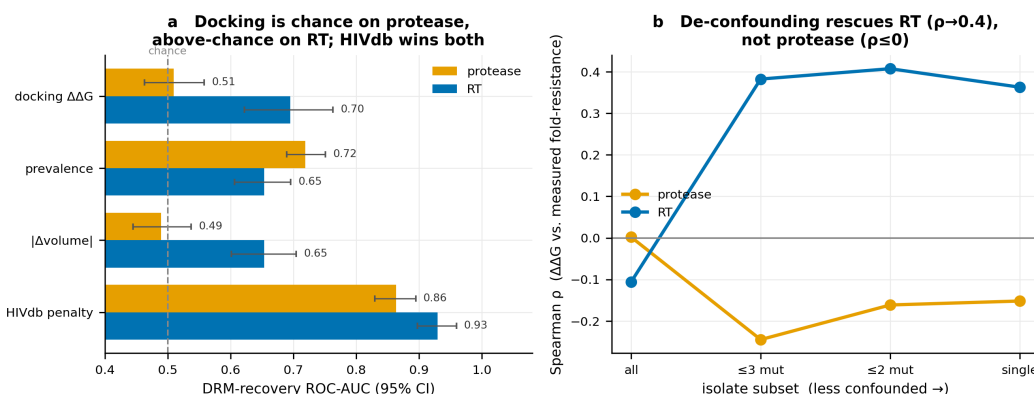


Figure 2: **Docking $\Delta\Delta G$ across the two targets.** (a) DRM-recovery ROC-AUC with bootstrap 95% CIs: docking- $\Delta\Delta G$ sits on the chance line for protease (0.51) and above it for RT (0.70), but a docking-free ligand-distance baseline beats it on both (Table 6), and the HIVdb expert system wins both. (b) De-confounding: as the fold-resistance target is rebuilt from progressively less co-mutated isolates, RT’s magnitude correlation climbs to $\rho \approx 0.4$ while protease stays ≤ 0 .

B Attribution ablation figures

C Per-drug breakdown and precision@ k

Table 2 gives the per-drug metrics behind the pooled numbers in §4.1, and Table 3 the decision-relevant precision@ k . Both are produced by `scripts/14_precision_at_k.py`.

Precision@ k ranks a *single* drug’s own mutation panel by predicted $\Delta\Delta G$ (descending, matching the pooled enrichment convention) and reports how many of the top k are known major DRMs. Significance is an exact one-sided hypergeometric test, giving the probability that drawing k mutations

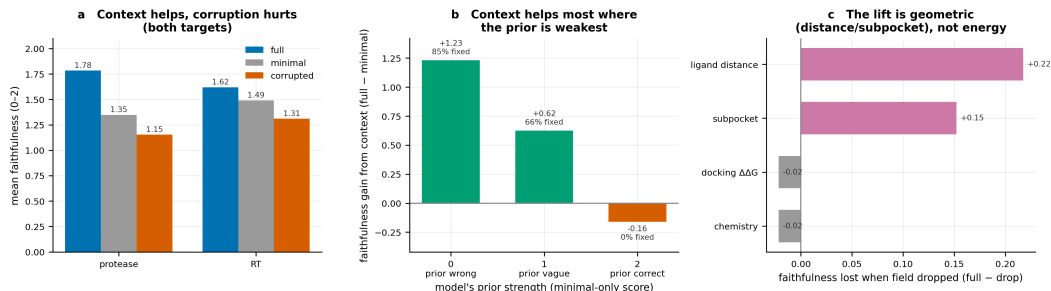


Figure 3: **Attribution ablation.** (a) Faithfulness under full / minimal / corrupted context, both targets: context helps, corruption hurts. (b) The benefit of correct context (full – minimal) by the model’s prior strength: largest where the prior alone is wrong (11 of 13 such cases corrected), zero where the prior is already correct. (c) Field-level ablation: dropping ligand distance costs most and subpocket next (both marginal at $n = 46$), while dropping the docking energy or the physicochemistry costs nothing measurable.

at random from that drug’s panel yields at least as many DRMs. That is appropriate at $k = 5$ where a bootstrap CI is uninformative.

The per-drug view shows the pooled result is not an artifact of averaging. Every NNRTI clears its base rate by a wide margin at every k ; among the PIs only atazanavir ($k = 5$) and lopinavir ($k = 20$) reach $p < 0.05$, and indinavir, whose ROC-AUC of 0.388 is *below* chance, returns no DRMs in its top 10. The protease failure is therefore not uniform noise: it is one target on which the method is, for some drugs, actively anti-correlated with the truth.

The last column is the comparison §4.1 rests on, and it runs opposite to the ROC-AUC ordering of Table 6. Distance is the better global ranker on both targets, but at $k = 10$ it wins only on protease (7 to 10 of 10, against 0 to 3 for docking) and loses on every NNRTI (3 to 4, against 7 to 8). The two metrics are measuring different things: distance orders the whole list well because DRMs are pocket-proximal, and then fills its top ten with residues that are close but not resistance mutations. Discriminating inside the pocket is what the energy contributes, and it contributes it only where the pocket is rigid.

Table 2: **Per-drug metrics.** DRM-recovery ROC-AUC and magnitude Spearman ρ vs. measured fold-resistance, per drug. n is the drug’s docked panel size, n_{DRM} its number of major DRMs.

target	drug	n	n_{DRM}	base rate	ROC-AUC	ρ
protease	ATV	258	33	0.128	0.533	−0.058
protease	DRV	209	30	0.144	0.462	0.033
protease	IDV	282	33	0.117	0.388	−0.085
protease	LPV	278	33	0.119	0.552	−0.013
protease	NFV	284	33	0.116	0.615	0.009
protease	SQV	280	33	0.118	0.672	0.138*
RT	EFV	423	30	0.071	0.738	0.078
RT	ETR	301	30	0.100	0.589	−0.003
RT	NVP	421	29	0.069	0.769	0.188**
RT	RPV	124	25	0.202	0.634	0.126

* $p < 0.05$, ** $p < 0.001$ for the Spearman correlation.

For completeness, Table 4 gives precision@ k on the *pooled* ranking (one ordering over all drug–mutation pairs of a target). The pooled protease view is the weaker of the two framings: only $k = 20$ clears $p < 0.05$, and pooled precision@10 is 2/10 against a 0.123 base rate. This is the same tension noted in §4.1, where protease shows tail enrichment without a usable ranking, and it is why we report the per-drug view as the primary one.

Table 3: **Per-drug precision@ k** , reported as (DRMs recovered)/ k . * marks $p < 0.05$ by an exact one-sided hypergeometric test against the drug’s own base rate.

drug	docking $\Delta\Delta G'$			distance	base rate
	$k = 5$	$k = 10$	$k = 20$	$k = 10$	
<i>protease</i>					
ATV	3/5*	3/10	3/20	8/10	0.13
DRV	1/5	1/10	2/20	10/10	0.14
IDV	0/5	0/10	1/20	7/10	0.12
LPV	2/5	3/10	6/20*	8/10	0.12
NFV	2/5	2/10	4/20	7/10	0.12
SQV	1/5	2/10	5/20	7/10	0.12
<i>RT</i>					
EFV	4/5*	8/10*	15/20*	3/10	0.07
ETR	3/5*	7/10*	11/20*	3/10	0.10
NVP	4/5*	8/10*	15/20*	3/10	0.07
RPV	3/5	7/10*	9/20*	4/10	0.20

Table 4: **Pooled precision@ k** , one ranking over all pairs per target. * $p < 0.05$ (exact hypergeometric).

target	$k = 5$	$k = 10$	$k = 20$	base rate
protease	2/5	2/10	6/20*	0.123
RT	4/5*	8/10*	14/20*	0.090

Table 5 is the full version of Table 1 with the bootstrap 95% CIs omitted there for width. The intervals show that protease docking’s ROC-AUC contains 0.5 while RT’s excludes it, and that both HIVdb intervals sit far above every other predictor on both targets.

Table 5: **Baselines with bootstrap 95% CIs** (5,000 resamples, seed 0). DRM-recovery ROC-AUC and magnitude Spearman ρ vs. measured fold-resistance. HIVdb’s ROC is partly circular against the expert DRM labels, so ρ is the fair column.

target	predictor	ROC-AUC [95% CI]	ρ [95% CI]
protease	docking $\Delta\Delta G$	0.510 [0.465, 0.557]	0.002 [−0.048, 0.052]
protease	mutation prevalence	0.719 [0.686, 0.751]	−0.078 [−0.127, −0.029]
protease	$ \Delta\text{volume} $	0.489 [0.445, 0.537]	0.142 [0.093, 0.191]
protease	HIVdb penalty	0.863 [0.828, 0.897]	0.360 [0.321, 0.395]
RT	docking $\Delta\Delta G$	0.695 [0.625, 0.768]	−0.106 [−0.164, −0.045]
RT	mutation prevalence	0.653 [0.608, 0.698]	0.127 [0.072, 0.182]
RT	$ \Delta\text{volume} $	0.653 [0.603, 0.701]	−0.009 [−0.065, 0.048]
RT	HIVdb penalty	0.929 [0.896, 0.959]	0.267 [0.209, 0.322]

Table 6 is the sharpest negative result in the paper: as a global ranker, the single number “how far is this residue from the ligand” outranks a full docking sweep on both targets, and on protease it also beats the expert system on the magnitude column. (At the top of the list the ordering reverses on RT; see Table 3.) Two readings are available and we do not think the evidence separates them. The deflationary one is that DRM status for a competitive inhibitor is close to a statement about pocket proximity, so a distance predictor is partly recovering the definition of the label; this is why we lead with the magnitude correlation against *measured* fold-resistance, where the same ordering holds but the margin is smaller. The constructive one is that the geometric localization of a mutation, which docking computes en route to its energy, is where the usable signal lives, and the scoring step discards rather than adds. The same conclusion arrives independently from the field-level ablation on the explanation side (§4.3), which is what makes us take it seriously.

Table 6: **Geometry-only predictors vs. docking** on the same scored pairs, with bootstrap 95% CIs (scripts/15_geometry_baseline.py). “-min distance” ranks a mutation by how close its residue lies to the ligand; “contacts ligand” is the binary $< 4 \text{ \AA}$ flag. Neither requires a docking run.

target	predictor	ROC-AUC [95% CI]	magnitude ρ
protease	docking $\Delta\Delta G$	0.510 [0.466, 0.557]	0.002
protease	-min distance to ligand	0.963 [0.954, 0.971]	0.387
protease	contacts ligand (binary)	0.776 [0.741, 0.809]	0.134
protease	$ \Delta\text{volume} $	0.489 [0.442, 0.535]	0.142
RT	docking $\Delta\Delta G$	0.695 [0.622, 0.764]	-0.106
RT	-min distance to ligand	0.800 [0.738, 0.854]	0.130
RT	contacts ligand (binary)	0.660 [0.615, 0.705]	0.178
RT	$ \Delta\text{volume} $	0.653 [0.604, 0.702]	-0.009

330 The leave-one-drug-out jackknife quoted in §4.1 refits the pooled ROC-AUC with each drug
 331 held out in turn: protease stays within [0.50, 0.53] and RT within [0.67, 0.74], so neither the
 332 protease null nor the RT signal is carried by any single compound. Both are produced by
 333 scripts/12_baselines.py.

334 D Worked example: one mutation through the ablation

335 To make the attribution ablation concrete, we trace a single pair, indinavir (IDV) with the protease
 336 flap mutation **M46I**, through all three context conditions. M46I is the paradigm weak-prior case: an
 337 *accessory* mutation that does not touch the inhibitor.

338 **Ground truth** (IAS-USA 2022): “*Met46 is a flap residue that does not contact the inhibitor directly.*
 339 *M46I is an accessory flap mutation that alters flap dynamics and closure and compensates for active-*
 340 *site mutations, broadly contributing to PI resistance.*”

341 Table 7 shows the context each condition receives.

Table 7: **Structural context supplied to the model** for IDV/M46I under each condition. *Corrupted* keeps the intrinsic chemistry (volume, hydrophobicity) but deranges the pipeline-derived geometry and energy across pairs.

field	full (real)	minimal	corrupted
min. distance to ligand ($\text{\AA}$)	7.0	—	4.1
distance to ligand centroid ($\text{\AA}$)	14.2	—	10.7
contacts ligand directly	False	—	True
region	flap	—	S2/S2' subpocket
Δvolume ($\text{\AA}^3$)	+3.8	—	+3.8
$\Delta\text{hydrophobicity}$	+2.6	—	+2.6
docking $\Delta\Delta G$ (kcal/mol)	-0.24	—	+0.66
judge score	2	0	0

342 **Full context** (score 2). “*The M46I mutation increases hydrophobicity while maintaining nearly*
 343 *identical volume, subtly altering the hydrophobic landscape of the flap region 7 $\text{\AA}$ from indinavir*
 344 *without directly contacting it... reflects reduced entropic penalty for flap closure and weakened*
 345 *indirect stabilization of inhibitor binding geometry, characteristic of flap mutations that achieve*
 346 *resistance through conformational rather than direct contact mechanisms.*” The judge scored 2: the
 347 correct primary mechanism.

348 **Minimal context** (score 0). “*The M46I substitution replaces a larger, flexible methionine with a*
 349 *smaller isoleucine, creating a structural cavity in the S1 subpocket where indinavir’s P1 ligand*
 350 *group normally makes critical van der Waals contacts...*” With only the drug and mutation name,
 351 the model defaults to an active-site packing story and *contradicts* the documented mechanism.

352 **Corrupted context** (score 0). “... isoleucine’s shorter branched aliphatic chain cannot maintain
 353 the same contact surface... the $\sim 4\text{\AA}$ minimum distance suggests the residue is in direct contact
 354 with the ligand... destabilizes the inhibitor by approximately 0.66 kcal/mol, particularly affecting
 355 the packing interactions critical for IDV’s potency in this region.”

356 The corrupted output is the informative one: the explanation *quotes the falsified values back* (the 4 $\text{\AA}$
 357 distance, the S2/S2’ region, the 0.66 kcal/mol energy) and reaches the wrong mechanism because
 358 of them. This is direct evidence that the generated mechanism is conditioned on the pipeline’s
 359 structural fields rather than retrieved from the model’s prior on a famous mutation name, and it is
 360 the single-instance version of the aggregate result in §4.3.

361 **Judge rationales.** The judge returns a one-sentence justification with every score; for this pair they
 362 are, verbatim:

- 363 • *full* (2): “The explanation correctly identifies M46I as a flap mutation that operates through
 364 conformational dynamics rather than direct contact, accurately describing how altered hy-
 365 drophobic landscape and flap closure dynamics indirectly weaken inhibitor stabilization,
 366 the core known mechanism.”
- 367 • *minimal* (0): “The explanation incorrectly claims M46I creates a structural cavity that dis-
 368 rupts direct van der Waals contacts with indinavir’s P1 ligand group, contradicting the
 369 known mechanism that Met46 is a flap residue that does NOT contact the inhibitor di-
 370 rectly...”
- 371 • *corrupted* (0): “The explanation contradicts the known mechanism by claiming M46I re-
 372 duces binding through direct van der Waals contacts with indinavir, when the ground truth
 373 explicitly states Met46 is a flap residue that does not contact the inhibitor directly...”

374 Both failures turn on the same discriminating fact, whether residue 46 contacts the ligand, which is
 375 exactly the field the corruption flips (`contacts_ligand_directly`: False $\rightarrow$ True) and exactly
 376 the field group the field-level ablation identifies as most load-bearing (§4.3, *drop_distance*). The
 377 same pair appears in all four PIs that carry M46I (ATV, IDV, LPV, NFV) with the identical 2/0/0
 378 score pattern, so it is a reproducible instance rather than a single sampled outlier.

379 E Reproducibility

380 **Structures and docking.** Protease: PDB 3OXC, chains A/B, catalytic Asp25 dyad asym-
 381 metrically protonated, mutations applied to both chains of the homodimer; box centre
 382 (5.341, -1.893, 14.179), size $22^3\text{\AA}$. RT: PDB 3V81, p66/p51 heterodimer, NNRTI pocket; box
 383 centre (41.105, 52.332, 49.098) (occupancy-weighted centroid of the chain-A nevirapine ligand;
 384 note that 3V81 contains a second NVP copy whose inclusion moves the centroid into empty space),
 385 size $24^3\text{\AA}$; wildtype absolute affinities of -9 to -12 kcal/mol confirm the box is on the pocket.
 386 AutoDock Vina 1.2 (CPU) / Uni-Dock (GPU), exhaustiveness 16, 5 poses retained per run, best
 387 pose scored. Receptors cleaned with PDBFixer; mutants built by point mutagenesis with a frozen-
 388 environment clash-relief minimization of the mutated side chain only; ligands prepared with RDKit
 389 (AddHs, ETKDG) and meeko. Crystal waters, including the conserved flap water, are stripped.

390 **Docking-dropout sensitivity.** Pairs whose docking returned no valid pose are excluded, which
 391 would bias the benchmark if those pairs were DRM-enriched. They are not: **zero** of the 97 excluded
 392 RT pairs and zero of the 6 excluded protease pairs are major DRMs, against a 9.0% DRM rate
 393 among the 1,269 scored RT pairs (Fisher exact $p = 3.6 \times 10^{-4}$). All 30 RT and all 33 protease
 394 primary DRMs survive into the scored set. The residual worry is the reverse: the excluded pairs are
 395 non-DRMs that, had they docked, might have scored high and acted as false positives. We bound
 396 that directly by imputing every excluded pair at the *worst case* (a $\Delta\Delta G$ above every scored pair, i.e.
 397 ranked top and all wrong): RT’s pooled ROC-AUC falls only from 0.695 to **0.641**, still above chance,
 398 and protease is unchanged (0.510 to 0.508). Median imputation moves neither target. Produced by
 399 `scripts/15_geometry_baseline.py`.

400 **Scale, yield and cost.** Protease: 1,597 drug-mutation dockings attempted across 6 PIs, of which 6
 401 (0.4%) returned no valid pose, leaving **1,591 scored pairs over 285 mutant receptors**. RT: 1,366
 402 attempted across 4 NNRTIs, of which 97 (7.1%) returned no valid pose, leaving **1,269 scored pairs**
 403 **over 423 mutant receptors**. Pairs without a valid pose are excluded from every analysis rather than

imputed; the RT failure rate is an order of magnitude higher than protease’s and is itself a caveat on the RT result, since the excluded set is not guaranteed to be missing at random. All 30 RT primary DRMs and all 33 protease primary DRMs are present in the scored set. The full RT sweep took 19 minutes on a single A100 80GB with Uni-Dock. At roughly 2.7 s per receptor this is orders of magnitude cheaper per mutation than an alchemical free-energy protocol such as Hauser et al. [11] (which we did not benchmark directly), and that cost/accuracy trade is the one the paper is about.

Models. Generation and judging both use `claude-haiku-4-5` (`max_tokens` 1024 for generation, 512 for judging); larger Claude models decline HIV-resistance *generation* but not judging, which is why the second judge (`claude-sonnet-5`) was available for the cross-model check. Scores are returned through a forced tool call against a fixed `report_score` schema, so every verdict is machine-readable and no free-text parsing is involved.

Generation prompt (verbatim system prompt; the enzyme noun and binding site are substituted per target, either “HIV-1 protease”/“active site” or “HIV-1 reverse transcriptase”/“allosteric NNRTI-binding pocket”): “*You are a structural biologist specializing in HIV-1 protease drug resistance. Given structural context about a point mutation and its effect on inhibitor binding, provide a 2-3 sentence mechanistic hypothesis for WHY this mutation causes resistance to this specific drug. Ground your explanation in the structural facts provided (residue size/charge change, distance to the active site, subpocket). Be specific about molecular interactions (van der Waals contacts, hydrogen bonds, steric clashes, electrostatic changes). Do not speculate beyond the given data, and do not restate the numbers back — interpret them. No preamble.*”

The user message is templated as

```
Drug: {drug_full} ({drug}), a {drug_class}
Mutation: {mutation}
Predicted delta-delta-G: {ddg} kcal/mol (positive = weaker binding =
resistance)
Structural context:
{context JSON, indent 2}

Explain the structural mechanism by which {mutation} likely reduces
{drug_full} binding.
```

The *minimal* condition drops the $\Delta\Delta G$ line and the context JSON entirely, keeping the drug and mutation lines and the closing instruction; *full* and *corrupted* reproduce the production prompt verbatim (with `cite=False`), so the ablation’s control arm is the real pipeline rather than a re-implementation.

Corruption procedure. The corrupted condition replaces six pipeline-derived fields with those of another pair: `distance_from_ligand_centroid_angstrom`, `min_distance_to_ligand_angstrom`, `contacts_ligand_directly`, `region`, `delta_delta_g` and `delta_g` (the $\Delta\Delta G$ shown in the prompt follows the donor). Donors are assigned by a *derangement* (sampled with seed 0, rejecting any permutation with a fixed point), so no pair keeps its own values and every corrupted context is realistic-but-wrong. Intrinsic-chemistry fields (position, wildtype and mutant residue identity, and the volume/charge/hydrophobicity changes) are deliberately *not* corrupted, because the mutation string itself determines them and scrambling them would make the context self-contradictory rather than merely false. A consequence worth stating: on low-cardinality fields such as `region` a derangement can coincide with the true value by chance, so the corruption is a lower bound on the effect a directed (antipodal) corruption would produce.

Contact-preserving corruption (control). Because `contacts_ligand_directly` is close to the answer for a mechanism-classification question, we ran a stricter variant that isolates the rest of the geometry. Donors are drawn *within contact class*, so contacting residues donate only to contacting residues, and the flag is restored to its true value afterwards, so it is both correct and consistent with the donated distance (no self-contradictory prompt). The derangement is over distinct *mutations* rather than (drug, mutation) pairs, since the same mutation recurs across drugs with identical geometry; the contact flag is preserved on all 46 protease pairs and the distance genuinely changes on 41 of them.

Faithfulness under this condition is 1.630 (72% correct), against 1.152 (41%) for the full derangement and 1.783 (80%) for real context. Three paired Wilcoxon contrasts follow: contact-preserved

beats full corruption by $\Delta = +0.478$ ($p = 5.8 \times 10^{-4}$), while real context still beats contact-preserved by $\Delta = +0.152$ ($p = 0.035$; 9 up, 3 down, 34 tied). The decomposition is therefore about 76% of the corruption effect attributable to the contact boolean and about 24% to distance, subpocket and energy. We report this because it materially qualifies §4.3: the pipeline does contribute beyond a single flag, but the headline *corrupted* arm overstates that contribution by roughly a factor of four. Produced by `scripts/17_contact_preserving_corruption.py`.

Field-level ablation groups. Each condition strips one group from the context: *drop_ΔΔG* removes `delta_delta_g` and `delta_g` and nulls the $\Delta\Delta G$ prompt line; *drop_distance* removes `distance_from_ligand_centroid_angstrom`, `min_distance_to_ligand_angstrom` and `contacts_ligand_directly`; *drop_subpocket* removes `region`; *drop_chem* removes `volume_change`, `volume_change_A3`, `charge_change` and `hydrophobicity_change`. All other fields and the prompt are unchanged.

Judge prompt (verbatim system prompt): “*You are evaluating whether an AI-generated explanation of HIV-1 protease drug resistance correctly identifies the known structural mechanism. Score 0 if the explanation contradicts the known mechanism, 1 if it is consistent but vague or misses the key interaction, 2 if it correctly identifies the primary structural basis. Judge only structural/mechanistic agreement, not writing style, and give a one-sentence justification.*” The enzyme noun is substituted per target. The judge sees the mutation, the drug, the ground-truth mechanism and the explanation. It never sees the condition label, the $\Delta\Delta G$, or which system produced the explanation.

Statistics. Permutation tests use 5,000 label shuffles and bootstrap CIs 5,000 resamples, seed 0. Enrichment ranks by $\Delta\Delta G$ descending. Paired ablation contrasts use the one-sided Wilcoxon signed-rank test at the (drug, mutation) level, which is the appropriate paired non-parametric test for the bounded, tied, non-Gaussian 0/1/2 score differences we observe; a paired *t*-test would assume a normality the difference distribution violates. Precision@*k* significance is the exact hypergeometric test described in Appendix C. We report many tests without multiplicity correction; the headline contrasts survive at $p < 10^{-3}$.

Ground truth. Protease mechanisms are hand-curated from the IAS-USA 2022 mutation list and the primary structural literature, with the source recorded per entry. RT mechanisms (22 entries) are built by a tool-using agent that queries live PubMed for each mutation, cites the papers it read, and self-verifies that the stated mechanism is supported before the entry is admitted; entries failing that check are dropped rather than included unverified. Because all RT ground truth is agent-built while all protease ground truth is human-curated, absolute faithfulness percentages are not strictly comparable *across* targets. The within-target ablation contrasts, which hold ground truth fixed, are.

Software. Python 3.11; pandas 2.2.3, numpy 1.26.4, scipy 1.15.3, scikit-learn 1.7.0, anthropic 0.74.0; RDKit, OpenMM, PDBFixer, AutoDock Vina and meeko via conda (`environment.yml`). The pipeline is reproduced by `scripts/01–14` in order; each number in this paper is produced by a named script, and the mapping is in the repository README.

Data availability. All docking results, explanations under every ablation condition, ground-truth mechanisms, judge scores with justifications, and the derived tables in this appendix are released with the code, so every figure and table here can be regenerated without re-running docking or re-querying any model.

F Failure-mode analysis

The method makes one implicit assumption: *that resistance is a local, contact-mediated perturbation of ligand binding, visible in a single rigid structure*. Every failure we observe violates one clause of that sentence, and which clause it violates predicts the failure. Table 8 organises them; the point is that these are not scattered noise but four identifiable regimes, three of which are properties of the *pipeline*, not of the model.

Two consequences worth stating plainly. First, eight of the nine faithfulness misses are attributable to what the pipeline could not represent rather than to the language model; only A98G is a plain model error. The E138 cluster is the clearest case: the LLM was faithful to its input and the input was mechanistically misleading. This is the sharpest illustration of the paper’s own thesis, since geometric context is what carries the explanation, so wrong or incomplete geometry is what breaks it. It also names a concrete fix (an inter-subunit contact feature) rather than a vague one. Second,

Table 8: **Failure modes**, grouped by the assumption each violates. Faithfulness misses are every score-0 explanation on both targets (3/46 protease, 6/55 RT).

case	assumption violated	mechanism of the failure
Protease active site	locality (resistance is a co-evolving network)	Substrate-envelope-constrained, multi-mutation resistance [13, 16]. Pooled ROC-AUC 0.51; indinavir is <i>below</i> chance at 0.388 with 0/10 precision@10. Rigid single-structure pocket affinity has no representation of the network.
L90M (all 3 protease score-0 misses: ATV, IDV, SQV)	contact-mediated	L90M does not contact the inhibitor; its documented mechanism is indirect distortion of active-site geometry. The model instead invokes dimer-interface destabilization or direct hydrophobic-contact loss. All three misses are the <i>same</i> mutation across three drugs.
E138A/G/K (4 of 6 RT score-0 misses)	contact-mediated & single-subunit	The true mechanism is the E138-K101 salt bridge across the p51/p66 interface [8]. The pipeline reports <code>min_distance_to_ligand</code> = 43.0 Å (measured to the chain-A ligand) and <code>contacts_ligand_directly</code> = False, so the model reasons <i>correctly</i> from a numerically true fact to the wrong conclusion, “~43 Å away, negligible impact.” The context has no inter-subunit coupling feature.
A98G (RT)	none (model error)	The context is correct and sufficient; the model simply dis-misses a real accessory contribution as “essentially neutral.” The only miss of the nine attributable to the model rather than the pipeline.
L100I (RT)	single rigid pose	The model attributes resistance to a change of binding mode; a single-pose rigid protocol cannot represent one.

the protease and RT failures are complementary: protease fails because resistance is non-local, RT’s residual failures because the locality *measure* is incomplete. Neither is repaired by a better language model.

This analysis is also the one part of the faithfulness evaluation that does not depend on an LLM. Each diagnosis above is checkable against artifacts outside the loop: that E138 sits 43.0 Å from the chain-A ligand is a fact of the structure file, and that Met46 and Leu90 are non-contact residues is recorded in the IAS-USA list. It is a partial check only, validating *why* the misses occur, not the absolute faithfulness percentage, but it does establish that the score-0 cases are real failures with identifiable causes rather than judge artifacts.

G An LLM-free structural-consistency check

Every check in §4.2 is internal to a family of language models. This appendix adds one that is not, by exploiting the fact that a single claim in each explanation is mechanically verifiable against the receptor: *does the mutated residue touch the ligand?*

Method. For each of the 303 explanations (46 protease and 55 RT pairs $\times$ 3 conditions) we take the true minimum residue–ligand distance from the structure (contact iff < 4 Å) and, separately, classify what the explanation *asserts* using a fixed regex lexicon in which negation is resolved first, so that “without directly contacting it” is scored as a non-contact claim rather than a contact one. Explanations making no explicit claim *abstain* and are excluded from accuracy rather than counted wrong. No language model participates in the measurement; the lexicon and the per-explanation calls are released with the code (`scripts/16_structural_consistency.py`).

Coverage. 139 of 303 explanations (46%) make an explicit contact claim. Abstention is itself condition-dependent: stripped of the distance field, the *minimal* arm makes far fewer checkable claims ($n = 14$), which is why that cell is the weakest of the three.

Result A: the ablation reproduces without an LLM (Table 9). Explanations written with the real structural context get the contact question right 88% of the time; stripped of it, 57%; given deranged geometry, 55% (Fisher exact $p = 0.013$ full vs. minimal, $p = 5.7 \times 10^{-5}$ full vs. corrupted).

The ordering, the magnitude and the near-equality of the minimal and corrupted arms all match the judge-scored result of §4.3, obtained by an entirely different route.

Result B: the judge tracks an external criterion. Explanations that are structurally *wrong* by this test receive a mean judge score of 1.02, versus 1.51 for structurally correct ones (Mann–Whitney one-sided $p = 6.1 \times 10^{-4}$; $n = 42$ and 97). The LLM judge is therefore not merely agreeing with other LLMs: its scores move with a criterion computed from a PDB file.

Table 9: **Contact-claim accuracy by condition**, computed from receptor geometry with no language model. Accuracy is over explanations making an explicit claim; abstentions are excluded.

condition	n claims	accuracy	protease / RT
full (real context)	60	0.883	0.862 / 0.903
minimal (no context)	14	0.571	0.500 / 0.667
corrupted (deranged)	65	0.554	0.583 / 0.537

What this does and does not establish. It is a check on one axis, over the 46% of explanations that commit to it, using a lexicon we wrote; it is not a substitute for expert grading of the full mechanistic claim, and we do not report it as one. What it does establish is that the two results most exposed to the circularity objection, the ablation contrast and the judge’s discrimination, both survive a measurement taken outside the LLM family.

H What each baseline needs to exist

Table 1 shows HIVdb beating docking on both targets. The argument for running docking anyway rests on *what data each predictor requires*, which Table 10 makes explicit rather than asserting.

Table 10: **Information each predictor requires** for a given (compound, target). “Novel compound” asks whether the predictor can be computed for a scaffold with no clinical history.

predictor	receptor structure	clinical isolates	expert curation	measured phenotypes	novel compound?
docking $\Delta\Delta G$	✓	—	—	—	yes
$ \Delta \text{ side-chain volume} $	—	—	—	—	yes
mutation prevalence	—	✓	—	—	no
Stanford HIVdb penalty	—	✓	✓	✓	no
supervised ML [3, 20, 4, 10]	—	✓	—	✓	no

Prevalence needs a corpus of clinical isolates genotyped *and* tested against that drug; HIVdb needs, in addition, expert curation of per-mutation per-drug penalties distilled from accumulated genotype–phenotype and outcome data; supervised models need labelled phenotypes for the drug they predict. None of these exist for a compound that has never been dosed in a patient. Docking needs only a receptor structure and the candidate’s chemistry.

What “prospective” does and does not mean here. Our $\Delta\Delta G$ comes from a physics/empirical docking engine (AutoDock Vina, or Uni-Dock’s GPU reimplementations of the same scoring function), not a learned one. This matters for how the result should be evaluated. Deep docking and co-folding models [5] carry billions of parameters trained on experimental complexes, so a held-out test set is mandatory to show they learned structural biology rather than memorizing training proteins, which is exactly the generalization gap Buttenschoen et al. [7] document. Vina is not parameter-free either: its handful of scoring-term weights were fit once by regression on general protein–ligand crystal data. But that fit involved no resistance phenotype, no HIVdb record and nothing in this evaluation. The precise claim is therefore not “nothing is fit” but *nothing in the $\Delta\Delta G$ path is fit to resistance data*, which is what rules out leakage into this benchmark. Consequently there is no resistance-specific training set to hold out, and the per-drug numbers in Table 2 are not a proxy for prospective performance. For a new compound against these targets, they *are* it.

Two boundaries on that. First, the *mutation panel* is target-derived from clinical data: docking tells you how a compound fares against a mutation, not which mutations to test. So the method is prospective per compound but not yet per target, and validating a genuinely novel target needs

573 a resistance dataset that, by construction, a novel target lacks. Second, the pipeline’s one heavily
574 trained component is the LLM, for which the memorisation concern is real and live; the *minimal*
575 condition of §4.3 is precisely its control, measuring what the model produces from the drug and
576 mutation name alone.

577 **The honest qualification.** Being training-free is not by itself the virtue: $|\Delta\text{volume}|$ is also com-
578 putable for any novel compound, needing nothing but the mutation string, and on RT it reaches
579 ROC-AUC 0.653 [0.603, 0.701] against docking’s 0.695 [0.625, 0.768], intervals that overlap sub-
580 stantially (Table 5). On protease it is 0.489, at chance, as docking is. And the ligand-distance
581 baseline of Table 6, also structure-only and also training-free, outranks docking globally on both
582 targets. So the defensible claim is narrower than “docking is the only prospective option”: among
583 prospectively-available predictors, docking is the best we measured only at the top of the ranking
584 and only on RT (Table 3), and on the global metric it is not separated from a zero-cost physicochem-
585 ical proxy. Separating them, and testing transfer to a new target, are the experiments that would
586 complete the argument.